

# URBANVIEW

## Business Requirements Document

v1.6 — Build Specification

July 2026 | Prepared by Artech Digital | Client: Monmaks d.o.o.

|                     |                       |                                          |
|---------------------|-----------------------|------------------------------------------|
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|---------------------|-----------------------|------------------------------------------|

### 1. What We Are Building

UrbanView tells you what you can build on a piece of land, and whether it is worth building. Today that answer takes days: planning rules sit in PDFs across different institutions, cadastral data sits elsewhere, market prices sit with agencies. Someone has to assemble it by hand, and mostly nobody does.

The user opens a map of Podgorica and clicks a plot. The system identifies the parcel, works out which planning zone governs it, and shows what the adopted plan permits — land use, maximum floors, site coverage, floor area ratio. It then calculates the money: buildable floor area, construction cost, market value, profit, ROI. Under two seconds, no login. If they want more than the automated answer, they pay for an expert analysis of that site, delivered by email.

This build is 2D and informative. 3D, AR, subscriptions and scenario comparison are excluded and belong to later phases. What it has to prove: that developers and architects will use a map-based feasibility tool, and that some will pay for a deeper look.

#### The flow in plain language

- User opens urbanview.io. Map of Podgorica loads with planning zone boundaries visible. No account.
- User finds a site — address, map click, or parcel number.
- System identifies the cadastral parcel, the planned urban parcel, and the governing zone.
- Panel opens: identification, then planning parameters, then market data and ROI. All financials as ranges.
- User toggles layers, or keeps browsing for a better plot.
- User optionally orders the expert report: short form, one-off payment, emailed manually by an expert.

*Two entry states, both must work. Some users arrive with a site and want to know what it permits. Others have no site and are hunting the city for one worth buying. The first needs search; the second needs browsing.*

### 2. Features to Build

*Scope and cooperation model. The functional prototype budget is EUR 10,000; the next phase will be funded at a six-figure level. Because the feature set below has grown during review beyond what EUR 10,000 covers, the client leaves Artech Digital to choose the cooperation model: either pay-per-project as a subcontractor, with the delivered scope aligned to the budget, or an associated-partner arrangement, where Artech is paid partly in fees and partly in equity in exchange for delivering the larger scope. Which features land in the prototype versus the next phase follows from that choice.*

#### 2.1 Map and Layer Engine

2D interactive map of Podgorica with switchable vector layers. Planning zone boundaries are the core layer and the default view — the first thing a user should see is how the city divides into zoning plans, because each zone carries different constraints. All non-core layers toggle independently, and are selected by clicking their visual representation — a layer card or icon — rather than checkboxes, radio buttons or eye icons. Map features are selectable and data-bearing, not flat imagery. The mapping provider (Mapbox or OpenStreetMap) is to be discussed and chosen on cost and capability.

| Layer                                    | Content                                                               | Priority    |
|------------------------------------------|-----------------------------------------------------------------------|-------------|
| Base map                                 | Roads and geography. Provider to be decided — OpenStreetMap or Mapbox | MUST        |
| Cadastral parcels                        | Existing parcels from the cadastral website                           | MUST        |
| Urban planning zones                     | Zone boundaries from adopted plans, each linked to its parameter set  | MUST — core |
| Planned urban parcels                    | Future parcel geometry from the plan. Separate layer — see 2.2        | MUST        |
| Planned land use                         | Land use designation from the plan                                    | MUST        |
| Urban blocks                             | Block boundaries within zones                                         | SHOULD      |
| Public ownership                         | Publicly owned parcels                                                | MUST        |
| Parcels under restitution / legal burden | Parcels subject to restitution claims or other legal burdens          | MUST        |
| Planned traffic network                  | Planned roads and traffic infrastructure                              | MUST        |
| Planning parameter heatmaps              | Coverage, height, FAR, GFA across the city                            | MUST        |
| Market data heatmap                      | At least one — preferably saleable area / price                       | MUST        |

*Coverage is defined by data, not administrative boundary. Only areas with adopted planning documentation are covered; plans in progress are excluded until adopted. Where there is no data the area is simply not covered — no error message, no empty state. Public ownership, restitution / legal-burden parcels, planned traffic and planning-parameter heatmaps are now required for the prototype (client, 24 July), along with at least one market-data heatmap, preferably saleable area. This reverses the earlier scoping and expands the ingestion effort — see the note below.*

*Scope note: adding these layers materially increases the data-preparation load, since each requires its own source, georeferencing and (for heatmaps) a computed surface across the city. Against the EUR 10,000 prototype budget this is significant and feeds directly into the cooperation-model choice noted at the start of Section 2. The olive-grove protection layer from BRD v2 remains excluded until the official olive cadastre exists.*

## 2.2 Parcel Identification and the Existing-vs-Planned Problem

The most important technical requirement in this document. The parcel in the cadastre today and the parcel envisaged by the adopted plan are frequently not the same — land gets taken for roads, public space, reparcelling. The client's example: a plot recorded at 100 units of area is reduced to 75 by the plan. The developer builds on 75. In most cases the planned urban parcel differs from the original cadastral parcel in both geometry and surface area — and this is not common knowledge; typically only the profession is aware of it. Surfacing it clearly is a core value of the platform, not a detail.

| Rule              | Requirement                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------|
| Separate layers   | Cadastral and planned parcels stored, rendered and labelled separately. Never merged.                     |
| Show both         | Where they differ, the panel shows both areas and states which drives the calculation.                    |
| Calculation basis | All output calculated on the planned urban parcel area where one exists; cadastral area only as fallback. |
| Never silent      | The mismatch is surfaced to the user, not resolved quietly.                                               |

Identification fields on selection: parcel number, sub-number, cadastral municipality, street address, governing planning document, urban block reference.

*Cadastral municipality is not optional. One political municipality contains several cadastral municipalities and the same parcel number recurs across them. Without it, the wrong parcel can be returned.*

## 2.3 Information Panel

The panel updates dynamically based on what the user selects, and it shows different content for a planning zone than for an urban parcel. Clicking empty zone area gives the zone-level picture; clicking a specific parcel gives the full parameter and financial set.

*Note on zones: in Montenegro, zones are not official areas with defined boundaries. They are UrbanView's own internal city divisions, roughly corresponding to city quarters, which group related planning documents. A single zone will usually contain multiple planning documents.*

**When a planning zone is selected:**

| Field                        | Note                                                                    |
|------------------------------|-------------------------------------------------------------------------|
| Zone name                    | UrbanView's internal city division ( $\approx$ city quarter)            |
| Planning documents           | The list of planning documents grouped under the zone — usually several |
| Document status              | For each: adopted, in progress, superseded                              |
| General planning information | Zone-level summary                                                      |

**When an urban parcel is selected, two groups. The user's job is to match one against the other: what the constraints allow you to build, against what the market says the location is worth.**

| Group 1 — Planning parameters              | Group 2 — Market data                    |
|--------------------------------------------|------------------------------------------|
| Parcel ID                                  | Estimated land value                     |
| Planning document name                     | Estimated design and documentation costs |
| Land use designation                       | Estimated construction costs             |
| Urban block reference                      | Estimated market value                   |
| Urban parcel number                        | Estimated saleable area                  |
| Urban parcel area                          | Estimated potential profit               |
| Maximum building height / number of floors | Estimated Return on Investment (ROI)     |
| Maximum site coverage % (IZ)               |                                          |
| Maximum Floor Area Ratio (II)              |                                          |
| Calculated maximum Gross Floor Area (GFA)  |                                          |

*Every planning value carries its source document reference. Parcel ID is UrbanView's own numeric unique identifier (e.g. 123456), part of a product strategy in which every element of the urban fabric — each parcel, building, and so on — has its own identity. It is not composed of the parcel number, sub-number and cadastral municipality; those are separate attributes displayed alongside it. All urban planning parameters and constraints are based on the planned urban parcels defined in the planning documents, not on the existing cadastral parcels. Estimated saleable area is shown explicitly as an output, not only as an internal step in the profit formula.*

## 2.4 Feasibility Calculation Engine

User input is the location. Nothing else — every other input is retrieved by the system.

| Output                | Formula                                                                                                                  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Maximum GFA           | FAR x plot area                                                                                                          |
| Maximum coverage area | Site coverage (%) x plot area                                                                                            |
| Potential profit      | (Max GFA x 0.70 x market value per m <sup>2</sup> ) – (land value + design and documentation costs + construction costs) |
| ROI (%)               | Potential profit / (land value + design and documentation costs + construction costs) x 100                              |

**Ranges, not single numbers.** Every financial output is low / expected / high. The client's reasoning was commercial: prices fluctuate, and a precise-looking figure invites a user to hold the platform to a promised return.

**Assumptions visible.** 70% of GFA assumed saleable for residential — indicative, varies by typology. Shown, not buried.

**Client owns the formulas** and validates every calculation before launch.

**Market data source (confirmed):** publicly available real estate listings (Realitica, Estitor) combined with Monstat official statistics for selling prices and construction cost indicators. The cadastral institution is not the market source — and the same applies to any other public authority responsible for preparing planning documents. Planning documents may include market values, but they carry lower weight than current real estate listings and Monstat statistics. Portals plus Monstat are the primary reference for land values, apartment sale prices and construction costs.

**User-editable assumptions (confirmed):** users can adjust assumptions such as construction cost, selling price and saleable-area percentage to test their own scenarios. The client considers this a valuable option. Edited values recalculate the ranges live; the deterministic formulas are unchanged.

*The engine is deterministic. AI does not generate the financial arithmetic — the client's words: "mathematically correct, not just AI assumptions." AI may interpret and normalise reference prices from listings and Monstat into the input values, but the calculation itself stays formula-driven.*

## 2.5 Data Sources and Preparation

All source data is publicly available official data. The challenge is not availability — it is that the data is scattered across sources and formats (PDF documents, GIS datasets, cadastral data, statistics) and has to be transformed into standardised, georeferenced GIS layers linked to the information panel.

| Source                                                                                                                                 | Provides                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| eRegistri — <a href="http://lamp.gov.me/PlanningDocument?m=PG">lamp.gov.me/PlanningDocument?m=PG</a>                                   | Adopted planning documents for Podgorica                                                                                                        |
| <a href="http://mondarchitects.com/site-check">mondarchitects.com/site-check</a>                                                       | Reference for the city's planning-document structure — the list of zones and the planning documents within each (select Podgorica municipality) |
| eKatastar — <a href="http://ekatastar.me">ekatastar.me</a>                                                                             | Ownership and legal burdens of cadastral parcels                                                                                                |
| eMapa — <a href="http://emapa.me">emapa.me</a>                                                                                         | Spatial and cadastral data                                                                                                                      |
| Geoportal Uprave za nekretnine (Real Estate Administration) — <a href="http://webmap.uzn.me/geoportal01">webmap.uzn.me/geoportal01</a> | Cadastral parcels                                                                                                                               |
| Realitica, Estitor                                                                                                                     | Real estate listings — market reference prices                                                                                                  |
| Monstat — <a href="http://monstat.org">monstat.org</a>                                                                                 | Official statistics: selling prices and construction cost indicators                                                                            |

*Representative planning documents and an instructions file are already in the shared Google Drive folder; further documents will be uploaded progressively during discovery.*

*The prototype has a second objective beyond validating the user workflow: establishing a repeatable data-preparation methodology that can be replicated across Montenegro and other countries. The full Montenegro application would cover 5–8 municipalities including Podgorica — more than 80% of the country's construction activity. Build the ingestion pipeline as a repeatable process, not a one-off Podgorica script.*

## 2.6 AI Document Extraction and Expert Approval

AI has one job: read the source documents so humans do not transcribe them. It handles automatic extraction of planning parameters from planning documents and AI-assisted interpretation of those documents, structures the parameters against the zone and block schema, and attaches a source reference to every value. It also interprets reference prices from real estate listings and Monstat data, using them as reference values for land prices, apartment sale prices and construction costs. Input data and some calculations may be prepared manually where that is faster. AI feeds reference values into the engine but does not perform the financial arithmetic, and does not publish anything unreviewed.

**Expert review:** 100% of AI-extracted planning information is reviewed and approved before publication. A prototype-stage control only — the client accepts it will not scale beyond one city and will be replaced by a sampling regime. Textual accuracy matters as much as numerical: a misread planning term changes the meaning of a constraint.

*Most planning source data is PDF — documents authored for print, not machines. Converting them into usable georeferenced geometry is the largest technical risk and cost in this build. The client noted on the call that a surveying or GIS engineer will probably be needed. Treat that as a probable additional cost, not an edge case.*

## 2.7 Paid Expert Analysis and Checkout

The prototype tests three revenue channels in parallel, to evaluate which model provides the strongest user value and willingness to pay: pay-per-service expert reports (one-off), a subscription for market data, and a subscription for full custom AI-assisted chat. The one-off expert analysis is specified below; the free/paid boundary that governs all three follows.

| Element (one-off expert analysis) | Specification                                                                                                                                                                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What is sold                      | Expert-led site and feasibility analysis of a specific location                                                                                                                                                                                                            |
| Price                             | EUR 50–200 per site. Band, not yet resolved into tiers, to be confirmed                                                                                                                                                                                                    |
| Billing                           | Single one-off transaction for the expert analysis. (Market data and AI chat are separate subscription channels — see the boundary below)                                                                                                                                  |
| Production and delivery           | Produced manually by a human expert, outside the platform, and emailed to the customer. Later stages may train AI on the expert's report template and prior expert steps to assist preparation and data extraction — with a human expert always approving the final report |
| Account                           | None for the one-off analysis — guest checkout, no registration to buy or receive. The subscription channels require registration                                                                                                                                          |
| Order form                        | First name, email, telephone. Company details for business purchasers                                                                                                                                                                                                      |
| Payment                           | EUR. Provider to be verified — Stripe (lower fees) vs Paddle (Montenegro availability to confirm)                                                                                                                                                                          |

### Free / paid boundary (confirmed):

| Free                                                                     | Paid                                                                   |
|--------------------------------------------------------------------------|------------------------------------------------------------------------|
| Explore locations across the city and review their development potential | Expert-led site analysis and feasibility study (one-off)               |
| Urban planning parameters — what can be built on a specific parcel       | Market data — whether it is worth developing the parcel (subscription) |
| 3 AI-assisted chat interactions per session                              | Unlimited AI-assisted interactions (subscription)                      |

*This settles the earlier open question. Market data moves behind the paywall; planning parameters stay free as the acquisition hook. The AI assistant is freemium — three interactions per session free, unlimited on subscription (see 2.8).*

*What the paid report adds (client-confirmed): the expert-led report provides site analysis, expert interpretation of the planning documents (some planning provisions are ambiguous and require professional judgment rather than a straightforward reading), hidden risk assessment, development scenarios, and benchmarking — insights beyond the raw figures already shown from the source data. The CTA button explains this in the panel.*

## 2.8 AI Assistant (Freemium)

A chat-like AI assistant that helps users explore the available information. Where the paid expert report is human-produced and deep, the assistant is automated and conversational: it answers questions about a selected location, explains what a planning parameter means, and points the user to the source. It is freemium — three AI-assisted interactions per session are free as part of the acquisition hook; unlimited interactions are a paid subscription (one of the three revenue channels in 2.7).

- **Scope:** reads the source documents and input data and answers questions grounded in them — so it is limited to, and effectively trained on, UrbanView's own approved data: planning parameters, zone information, and the figures shown in the panel. Additional context documents may be provided to broaden it — laws, regulations, building codes, news — since the assistant needs slightly wider context than the parcel data alone. It does not invent planning values or generate the financial arithmetic.
- **Guardrails:** the same controls as the rest of the platform — it draws only on expert-approved data and cites the source document, and it does not present unreviewed extractions as fact.
- **Boundary:** the assistant helps users navigate and understand; it does not perform the expert interpretation, risk assessment or development-scenario work reserved for the paid report.

*New scope. The AI assistant is a feature the client added during review. It runs on a freemium model — three free interactions per session, unlimited on subscription. If built on a third-party service (Gemini, ChatGPT, DeepSeek, etc.) the free tier is cheap to run; a full custom AI solution is the paid subscription channel. Either way it introduces a conversational LLM surface over the platform's data and should be reflected in the estimate and timeline alongside the expanded layer set in Section 2.1.*

## 2.9 Admin Panel

Role-gated, separate from the public platform. This is where the data pipeline is operated.

- Upload and version planning documents, GIS files and cadastral extracts
- Review, amend and approve AI-extracted data before it publishes
- Create and edit zone parameter sets; manage which areas are live
- Update financial assumptions — construction rates, market benchmarks, range boundaries
- View and assign paid analysis orders

Audit logging of changes and approvals is unresolved. Recommendation: build it. Once a published value is wrong, the first question is who changed what and when.

## 2.10 Behavioural Data Capture

Selective capture of location queries, parcel selections and layer interactions, for later resale as location-demand insight. The client will specify which interactions matter. Personal data limited to what a purchaser submits. Privacy notice required at launch; legal basis for commercial reuse still to be confirmed.

## 3. Screen-by-Screen Build Spec

The map is always present; panels open over it rather than replacing it. No dead ends — every path, including an uncovered area or a failed payment, has a clear outcome.

| S1 | Map — Landing                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <p><b>UI:</b> Full-screen 2D map of Podgorica, planning zones visible and visually distinguished. Search bar top-centre, layer controls left. No login wall, no splash.</p> <p><b>Logic:</b> Loads at city extent. Only covered areas render zone geometry.</p> <p><b>Future phase:</b> Once multiple cities are covered, the user selects a city from the list of covered cities; the initial default view is the capital.</p> |

| S2 | Find a Location                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <p><b>Input:</b> Address with autocomplete, direct map click, or parcel number. Only one of the three is required to identify a location — the user picks whichever suits them.</p> <p><b>Logic:</b> Captures coordinates, resolves cadastral parcel, planned parcel and zone by point-in-polygon. Under 2 seconds.</p> <p><b>Note:</b> The parcel number is the official cadastral reference used for search; Parcel ID is UrbanView's internal identifier (see 2.3), not a search input. Client estimates roughly 90% of searches will be by address or map click combined; parcel number pinpoints an exact boundary when known.</p> <p><b>Next:</b> Resolved → S3. Outside coverage → S6.</p> |

| S3 | Information Panel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <p><b>UI:</b> Side panel opens right, selection highlighted on map. Content updates dynamically with what is selected.</p> <p><b>Zone selected:</b> Planning document name, planning zone name, document status, general planning information.</p> <p><b>Parcel selected — Group 1:</b> Parcel ID, planning document, land use, urban block, parcel number, parcel area, maximum height/floors, site coverage %, FAR, calculated maximum GFA. Every value carries a visible source reference.</p> <p><b>Parcel selected — Group 2:</b> Land value, design and documentation costs, construction costs, market value, saleable area, potential profit, ROI. Calculated automatically, no button. Every figure a range.</p> <p><b>Rules:</b> All planning parameters and constraints are based on the planned urban parcels defined in the planning documents, not on the existing cadastral parcels. Where cadastral and planned areas differ, both shown with the planned area stated as the calculation basis. Assumptions visible. One-sentence disclaimer that figures are indicative, not investment advice.</p> |

| S4 | Order Expert Analysis                                                                                                                                                                                                                                                                                                                                                                                     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <p><b>UI:</b> Call to action within the panel, explaining in one or two sentences what the report adds beyond what is on screen.</p> <p><b>Form:</b> First name, email, telephone. Optional company details. Short — the client specified "a tiny form, not a huge form." Location carried through automatically.</p> <p><b>Logic:</b> No account, no password, no verification step before purchase.</p> |

| S5 | Payment and Confirmation                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <p><b>UI:</b> Hosted checkout. Price stated before the user commits. One-off payment in EUR, no card data on UrbanView infrastructure.</p> <p><b>Failure:</b> Returns to S4 with form data preserved and a one-sentence explanation.</p> <p><b>Confirmation:</b> Order reference, location ordered, turnaround time. Same by email. Order lands in the admin queue; expert notified.</p> <p><b>Open:</b> Notification routing, queue ownership and committed turnaround still to be defined.</p> |

| S6 | Outside Coverage |
|----|------------------|
|----|------------------|

|  |                                                                                                                                                                                                                                                                                                                                     |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>UI:</b> No error message, no red state, no apology modal.</p> <p><b>Logic:</b> Zone geometry absent, panel does not open, base map stays navigable.</p> <p><b>Rationale:</b> Client decision: the platform covers only areas with adopted planning data. Uncovered areas are a fact of the data, not a failure to report.</p> |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 4. UX Rules — Non-Negotiable

| Rule                               | What it means in practice                                                                                                                                                                |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Guessable on first use             | The client's reference was an arcade game — a first-time user works it out by trying. No tutorial, no onboarding tour.                                                                   |
| The map is never replaced          | Panels open over the map. The user never loses spatial context by navigating away.                                                                                                       |
| Two groups, always separated       | Planning constraints and market data are visually distinct. The user's job is comparing them.                                                                                            |
| Ranges, never single figures       | A precise-looking number is a promise the platform cannot keep.                                                                                                                          |
| Every value cites its source       | Any planning parameter traceable to its document in one click. This is a trust product first.                                                                                            |
| No error state for uncovered areas | Absence of data is a fact, not a failure to report.                                                                                                                                      |
| Free tier is the hook              | Exploring locations, planning parameters, and three AI chats per session are free. Market data, the expert report and unlimited AI chat are paid. The boundary is unambiguous on screen. |
| No account to buy                  | Guest checkout. Registration is not a gate on revenue in this phase.                                                                                                                     |
| Under two seconds                  | Query to populated panel. Speed is the core claim — days to minutes.                                                                                                                     |
| Desktop-first                      | Web app, desktop browser first, functional on tablet and phone. No native app.                                                                                                           |

## 5. Technical Decisions

| Component                 | Decision                                                                               | Notes                                                                                                                                                                                                                             |
|---------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document extraction       | LLM extraction of text and tables from planning PDFs, with human approval queue        | Structured output, not prose. Source reference on every value.                                                                                                                                                                    |
| Geometry extraction       | To be determined by sample assessment                                                  | Vector PDF may allow direct extraction; scanned raster needs manual redraw. Largest unknown in the build.                                                                                                                         |
| Planning rule application | Rule-based, from the parameter table, with LLM assistance where input structure varies | Input data does not always follow the same structure or matching logic, so an LLM may be needed to map documents to the schema. Final applied values are still expert-approved.                                                   |
| Financial calculation     | Deterministic formula engine; LLM may assist in normalising and matching input values  | The client noted input data does not always follow the same structure or matching logic, so an LLM may help map source values into the engine. The arithmetic itself stays formula-driven and client-supplied — not AI-generated. |
| Map engine / base map     | Mapbox over OpenStreetMap                                                              | Vector layers with independent visibility control.                                                                                                                                                                                |
| Geocoding                 | OSM-based, with autocomplete                                                           | Address search suggestions.                                                                                                                                                                                                       |
| Cadastral parcels         | Cadastral website — method to confirm                                                  | API, export or scrape. Availability and licensing to verify.                                                                                                                                                                      |

| Component    | Decision                                   | Notes                                                                                                 |
|--------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Payments     | Stripe or Paddle, EUR, one-off             | Hosted checkout, no card data stored. Verify Stripe Montenegro availability — lower fees than Paddle. |
| Frontend     | React / Next.js                            | Map interface, panel, checkout, admin.                                                                |
| Backend      | Node.js or Python                          | Calculation engine, extraction orchestration, orders.                                                 |
| Database     | PostgreSQL + PostGIS                       | Zones, parcels, blocks, parameter sets. Version-controlled datasets.                                  |
| File storage | S3-compatible, private buckets             | Source documents, GIS files, generated reports.                                                       |
| Platform     | Web, desktop-first, mobile-responsive      | No native iOS or Android app in this build.                                                           |
| Ownership    | All accounts and repositories client-owned | Delivery team added as collaborators. Code is client property under contract.                         |

## 6. Additional Requirements

### 6.1 Legal and Compliance

- Terms of Service, Privacy Policy, Refund Policy and an indicative-figures disclaimer required at launch. Drafted by a lawyer familiar with Montenegrin consumer and data protection law
- Data licensing: the platform commercially displays data derived from public planning documents and the cadastre. Permission per source is outstanding — blocks commercial launch, not the build
- Liability: the platform publishes ROI figures people may act on. Ranges and visible assumptions mitigate this; a clear "indicative, not investment or planning advice" statement is required on every screen showing financials
- A paid, human-produced analysis carries professional exposure of a different order to an automated estimate. The expert's standing and any indemnity cover should be established before the first order

### 6.2 Analytics

Tracked from day one — the prototype is a validation instrument, and untracked usage proves nothing.

- map\_loaded, search\_performed, parcel\_selected, layer\_toggled
- panel\_viewed, financials\_viewed, source\_reference\_opened
- order\_started, checkout\_completed, return\_visit, sessions\_per\_user

Client dashboard: conversion at each funnel step, orders and revenue, most-searched districts, repeat usage rate.

### 6.3 Support and Content

- Email support to a shared inbox, 24 business hours. Short FAQ covering what the data means, where it comes from, what the paid analysis includes
- Planning terminology glossary — the platform uses terms (IZ, FAR/II, cadastral municipality, urban block) that its own users define inconsistently. Plain-language definitions are a trust feature
- Refund policy not yet defined. Recommendation: full refund before delivery, discretionary after. Needed before the first paying customer
- 3–5 example outputs for the landing page as the primary trust signal

## 7. How We Know It's Working

| Metric                   | Prototype target | MVP target | Source               |
|--------------------------|------------------|------------|----------------------|
| Pilot professional users | 20 minimum       | 100+       | Recruitment tracking |

| Metric                          | Prototype target        | MVP target            | Source                 |
|---------------------------------|-------------------------|-----------------------|------------------------|
| Sessions per pilot user         | 3+                      | 5+                    | Session analytics      |
| Query response time             | Under 2 seconds         | Under 1 second        | API monitoring         |
| Uptime, business hours          | 99%                     | 99.5%                 | Uptime monitoring      |
| Capacity                        | ~1,000 users            | 5,000+                | Load testing           |
| Coverage at launch              | Podgorica, full adopted | Podgorica, Bar, Budva | Admin coverage manager |
| Published value accuracy        | 100% expert-verified    | Sampling regime       | Review queue audit     |
| Paid analysis orders            | Demand demonstrated     | Sustained volume      | Payment provider       |
| Panel views reaching financials | Baseline established    | 60%+                  | Funnel tracking        |

*What to watch for. If users select parcels but never scroll to the financials, the panel is too long — or the planning group is answering their question on its own, which changes what the paid product should be.*

*If nobody orders the expert analysis, the problem is more likely the undefined value proposition (still undefined) than the price. Fix the description before discounting.*

*If professional users report wrong planning values, stop and audit the extraction pipeline before adding coverage. One wrong FAR shown to an architect costs more than three months of missing features.*

## 8. Post-Prototype Additions (Not in This Build)

**MVP:** expansion to Podgorica, Bar and Budva; parcel and scenario comparison; AI feasibility engine and ML models trained on planning documentation; 3D module with constraint-aware push-and-pull modelling; 3D city models and AR; project model upload and sharing for architects, chief city architects and developers; user accounts and subscription tiers with exportable reports, benchmarking and scenario comparison.

**Full Montenegro:** 5–8 municipalities including Podgorica, representing more than 80% of the country's construction activity. The prototype's data-preparation methodology (Section 2.5) is built to be replicated across these municipalities and, later, other countries.

**MMP:** Urban Modeler with prompt-based modelling and GIS import; full cadastral integration and City Information Model; advanced financial modelling and bill of quantities; automated compliance reporting; revenue from data insights, investment matching and advertising.

**Market expansion:** Western Balkans, then German-framework jurisdictions, then Anglo-Saxon. Pareto coverage per country — the ~20% of municipalities accounting for ~80% of construction. Plus the Heritwin module for heritage settlements outside planning coverage.

*Architectural implication for this build: isolate market-specific ingestion and planning terminology from the core engine. Adding a city or country should be a data and configuration exercise, not a rebuild. This is the one piece of future-proofing worth paying for now.*

## 9. Methodology Pipeline (User's step-by-step when analyzing the location/parcel)

In the following text, we will explain our steps that we use everytime we analyze a parcel for purposes of offering our services or when we have contract to design the object. The methodology consists of:

- 1. Step:** Searching for the specific cadastral parcel in the urban plans that also contain existing condition and planned condition. First check is if the cadastral parcel is the same geometry and area as planned one - urban parcel. During that research we check the ownership status of the parcel.
- 2. Step:** We look for parts from urban planning PDF's that contain specific information:

- a. Planned urban parcels - regulation and its relation to public areas
  - b. Planned urban regulation - Area envisaged for site coverage by the object, building height and number of floors, linear distance from the neighboring parcels and public areas.
  - c. Planned land use within the block or area that has the same markaton
  - d. Existing and planned infrastructure utilities - water, sewage, energy infrastructure, heating, access roads etc.
- 3. Step:** After analyzing the parcel and the referent PDF planning document, we start designing the orthogonal projection (2D view from the top) of the object to the parcel, implementing all the rules and regulations.
- 4. Step:** After the design of the object we start working on 3D view and we start fitting the object into the build environment.
- 5. Step:** This step is the final. After we conclude that it is the design that should be presented to the Client, we prepare preliminary 2D plans, 3D model and comprehensive analysis of real estate development venture.
- 6. Step:** If the Client decides to go for the venture, we prepare offer and start designing.

*A few items remain the last gate before wireframing and estimation can be finalised: the definitive Podgorica zone and planning-document list with parcel counts (a reference is now available at [mondarchitects.com/site-check](https://mondarchitects.com/site-check), to be confirmed against the official registry), the approved financial formulas, and payment provider verification (Stripe vs Paddle). The free/paid boundary is now settled (2.7). The expanded layer set in Section 2.1 and the three revenue channels also need to be reflected in the estimate and timeline. Everything else is stable.*